Lab 1: Orders of Magnitude and Parallax*

September 16, 2026

1 Introduction

Astronomy asks us to ponder structures and systems on scales way outside the realm of human experience. From itty-bitty atoms to massive clusters of galaxies, from the speed of light to the age of stars, the sheer extremity of the numbers we toss around sometimes allows them to lose their meaning. If I tell you that the Sun has a mass of 1,989,000,000,000,000,000,000,000,000,000 kg, this likely wouldn't mean much to you. However, a working understanding of numbers, units, and especially **orders of magnitude** is a vital tool of scientific literacy. In this lab, we will develop skills to rescue numbers from mathematical abstraction and root them in a familiar context (for example, comparing the Sun's mass to something you can actually picture).

After getting comfortable with working across large scales and orders of magnitude, we will learn how to calculate the *very large* distances we encounter in astronomy. Using basic concepts from trigonometry, we can figure out the distances of our neighbors (like the Sun and other stars in the Milky Way) with a method called **astronomical parallax**.

2 Scientific Notation & Orders of Magnitude

When writing out big numbers like the Sun's mass,

$$1,989,000,000,000,000,000,000,000,000,000 \text{ kg},$$

or really small numbers like the mass of an electron,

$$0.0000000000000000000000000000009109 \text{ kg},$$

it is useful to use **scientific notation** to avoid writing out so many zeros. We can do this with powers of 10: instead of writing 100,000 (1 followed by 5 zeros) we can write 10^5 ("10 to the power of 5" or "10 to the fifth power"). Likewise, we can write 0.0001 (1 preceded by 3 leading zeros or zeros in decimal places). The mass of the Sun can thus be written as 1.989×10^{30} kg and the mass of an electron as 9.109×10^{-31} kg, where leading numbers, such as 1.989 and 9.109 are referred to as **coefficients** and the powers of ten, 10^{30} and 10^{-31}, are **orders of magnitude**.

2.1 Significant Figures

Another convention of writing numbers in scientific notation is always having only *one* digit to the left of the decimal points; this standardizes the definition of "order of magnitude" and makes it easier to compare measurements expressed in scientific notation. If we can only have one digit to

*Adapted from materials developed by previous graduate teaching assistants in the Department of Astronomy.

Large Number Formats: $10^{13.4024}$, a person who has come back to numbers after a journey deep into some random theoretical field (`https://xkcd.com/2319/`).

the left of the decimal points, how about to the *right* of the decimal point? This is specified by the number of **significant figures** or "sig figs".

3.4578×10^9 has five sig figs, while 3.5×10^9 has two sig figs and 3×10^9 only has one sig fig. It's important to keep significant figures in mind when reporting measurements, since your measuring tool may not be precise enough to justify many sig figs – for example, if your ruler only has markings at each centimeter, it would be more proper to cite a measurement of 3.5 cm than to cite a measurement of 3.48825 cm (the ruler is definitely not precise enough to give you a measurement to six significant figures). Usually, keeping two or three significant figures is enough (and often more correct).

Qs: Let's try some problems involving scientific notation and orders of magnitude. **Record your answers in your lab write-up.**

1. The mass of the Earth is 5.97×10^{24} kg. What is the order of magnitude of this number? The mass of the Milky Way is about 6×10^{42} kg – by how many orders of magnitude is the mass of the Milky Way greater than the mass of the Earth? How many times do we need to multiply the mass of the Earth by 10 in order to reach the order of magnitude of the mass of the Milky Way?

2. The age of the Universe is thought to be around 13787×10^6 years. How many significant figures does this measurement have? Write this number in proper scientific notation using *three* significant figures. Write the number again in scientific notation using *two* significant figures. (hint: when reducing the number of sig figs, you'll need to *round* the original measurement to the appropriate number of decimal places)

3. Your friend is writing a scientific paper about her cat. In her manuscript, she reports her cat's mass to be 4.536782 kg. Why might you be skeptical of this measurement? What additional information (e.g., about the equipment used to make this measurement) could your friend include in her paper that would make this number more believable? Write the cat's mass using three significant figures.

2.2 Arithmetic with scientific notation

Scientific notation gives us a convenient way of compactly writing very big and very small numbers. By comparing orders of magnitude, scientific notation also gives us a quick and easy way to determine how big (or how small) one measurement is compared to another. We can also simplify arithmetic with scientific notation and numbers involving differing orders of magnitude. Here are some quick rules about arithmetic with scientific notation:

- When **adding** or **subtracting** two numbers in scientific notation, first match exponents/orders of magnitude and add or subtract the coefficients.

$$3.2 \times 10^5 + 4.5 \times 10^4 \rightarrow 3.2 \times 10^5 + 0.45 \times 10^5 = (3.2 + 0.45) \times 10^5 = \boxed{3.65 \times 10^5}$$

  Note: if exponents differ by 3 or more orders of magnitude, the smaller number barely affects the result, a good rule of thumb is to ignore it for a quick estimate (e.g., $6 \times 10^8 + 2 \times 10^5 \approx 6 \times 10^8$*)*

- When **multiplying** two numbers in scientific notation, *multiply the coefficients* but *add the magnitudes*. For example:

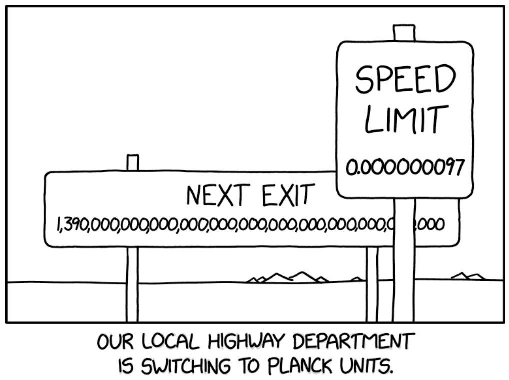
$$(3.4 \times 10^9) \times (5.8 \times 10^4) = (3.4 \times 5.8) \times 10^{9+4} = 19.72 \times 10^{13} = 1.972 \times 10^{14} = \boxed{2.0 \times 10^{14}}$$

  Notice that the product 19.72×10^{13} was not in proper scientific notation, so I had to slightly adjust my answer. I also rounded my final answer to two significant figures – usually, when multiplying or dividing numbers in scientific notation, it's proper to write your final answer using the *least* number of sig figs that appear in the original numbers (in this case, 3.4×10^9 and 5.8×10^4 each had two sig figs, so the least number of sig figs appearing in the original numbers was two. If, instead, we had started with 3×10^9 and 5.8×10^4, our final answer should have had only *one* sig fig).

- When **dividing** two numbers in scientific notation, we *divide* the coefficients but *subtract* the orders of magnitude. For example:

$$\frac{3.4 \times 10^9}{5.8 \times 10^4} = \frac{3.4}{5.8} \times 10^{9-4} = 0.586 \times 10^5 = 5.86 \times 10^4 = \boxed{5.9 \times 10^4}$$

  Again, I had to adjust my final answer to be in proper scientific notation, and I rounded to two significant figures to match the least number of sig figs appearing in the original numbers.



Highway Signs: Highway engineers like Planck units because, like the speed of light, the energy capacity of a typical car's gas tank is 1 (`https://xkcd.com/3289/`).

The sig fig rules ensure that our final answer is only as precise as our *least* precise measurement. If you forget these rules, it's usually fine to just keep your final answer to one or two sig figs – it's better to have too few significant figures (i.e., less precision) than too many.

Qs: Let's try some problems involving arithmetic using what we've learned. **Record your answers in your lab write-up.** Be sure to show all your intermediate steps in your arithmetic!

4. Light travels at a speed of 1.8×10^7 kilometers per minute. The average distance from the Earth to the Sun is 1.5×10^8 kilometers. Find the time (in minutes) that it takes light to travel from the Sun to the Earth by *dividing* the Earth-Sun distance by the speed of light. Give your answer in scientific notation with the proper number of significant figures. If the Sun were to suddenly stop emitting light, how long would it take us (on Earth) to notice this?

5. The speed at which the Earth orbits around the Sun depends on the *sum* of the Earth's mass and the Sun's mass. The mass of the Earth is 5.972×10^{24} kg and the mass of the Sun is 1.989×10^{30} kg. Add these two masses together and express your answer in scientific notation with the proper number of significant figures. Do you think the mass of the Earth play a significant role in determining the Earth's orbital speed around the Sun? Explain your answer.

3 Units and Unit Conversions

If someone tells you that the height of a tree is "6," you might be a little confused – is the height 6 feet? 6 meters? 6 inches? Whenever you report a measurement, you *must* cite the **units** – class is not "3" long, it's 3 *hours* long. Without units, a measurement doesn't have as much meaning, and it isn't as specific.

In the physical sciences, we often use the "International System" of units, or the "SI" system. In the SI system, the typical unit of mass is the kilogram (kg), the typical unit of length is the meter (m), and the typical unit of time is the second (s). These units are fine in many cases, but may not always be the most convenient – what if we instead wanted to measure time in years? Or distance in megaparsecs? Or mass in grams? In these cases, we must **convert** the units of our measurement, or perform a "unit conversion."

A nice trick for converting units is to multiply the original value by a "conversion factor" that's equal to one. For instance, since 365 days is equivalent to 1 year, the conversion factor $\frac{1\,\text{year}}{365\,\text{days}}$ is equal to one. So, if we wanted to convert a measurement of 30 days to units of years, we would do

$$30\,\text{days} = (30\,\text{days}) \times \left(\frac{1\,\text{year}}{365\,\text{days}}\right) = \frac{(30\,\text{days}) \times (1\,\text{year})}{365\,\text{days}} = \frac{30}{365}\,\text{years} = \boxed{8.2 \times 10^{-2}\,\text{years}}$$

Notice that, because "days" appears on both the top and the bottom of the fraction, that unit *cancels out*, just like if we were dividing an ordinary number by itself. It's very important that you always set up your conversion factor such that this cancellation occurs. Here's another example, where we convert 3×10^8 meters per second (m/s, a unit of speed or *velocity*) to meters per hour:

$$\begin{aligned} 3 \times 10^8\,\text{m/s} = \left(\frac{3 \times 10^8\,\text{m}}{1\,\text{s}}\right) \times \left(\frac{3600\,\text{s}}{1\,\text{hr}}\right) = \frac{(3 \times 10^8\,\text{m}) \times (3600\,\text{s})}{(1\,\text{s}) \times (1\,\text{hr})} &= \frac{3 \times 10^8 \times 3600}{1}\,\frac{\text{m}}{\text{hr}} \\ &= \frac{1 \times 10^{12}}{1}\,\frac{\text{m}}{\text{hr}} = \boxed{1 \times 10^{12}\,\text{m/hr}} \end{aligned}$$

Note how I wrote $3 \times 10^8\,\text{m/s}$ as a fraction, and then constructed the conversion factor $\frac{3600\,\text{s}}{1\,\text{hr}}$ such that the units of seconds (s) canceled out. When carrying out a unit conversion, I strongly recommend writing the units explicitly as fractions to help keep track of which units cancel and which units end up in your final answer. As a final example, let's convert 1.48 inches/year to feet/decade, which will require *two* conversion factors – here, I'll slash through the units that cancel out:

$$1.48\,\text{inches/year} = \frac{1.48\,\cancel{\text{inches}}}{1\,\cancel{\text{year}}} \times \frac{1\,\text{foot}}{12\,\cancel{\text{inches}}} \times \frac{10\,\cancel{\text{years}}}{1\,\text{decade}} = \frac{1.48 \times 10}{12}\,\frac{\text{feet}}{\text{decade}} = \boxed{1.23\,\text{feet/decade}}$$

Unit conversions can be easy to mess up, especially when you're working with some particularly strange ones, so a good rule-of-thumb to check your answers is trying to see if it makes physical sense. If you're calculating how tall a 6 ft tree would be in inches, it wouldn't make sense if your answer was less than 6! Since inches are smaller than feet, your answer will be much larger than 6.

Qs: Unit conversion can be tricky when you first start out, but practice makes perfect! To get some practice with unit conversions, let's do some problems. **Record your solutions in your lab write-up**. Be sure to show your work!

6. Find someone you don't know in the class. Ask them their name, and when their birthday is. How old are they? Record your answer here. You should have the day, month, and year.

7. Now, use unit conversion to figure out how many seconds old they are. Hint: There are 365 days in one year, 24 hours in a day, 60 minutes in an hour, and 60 seconds in one minute. You can use Google to see how many days it's been since their birthdate.

8. The distance between us and the Andromeda Galaxy is 7.7×10^5 pc (pc = parsec). 1 parsec is equivalent to 3.09×10^{13} kilometers. What is the distance to the Andromeda Galaxy in units of *kilometers*? 1 parsec is also equivalent to 3.26 ly (ly = light years). What is the distance to the Andromeda Galaxy in units of *light years*? How many kilometers are in a light year?

9. Based on your answer to the last question, what type of unit is a light year? Speed, distance, currency, mass, time, etc?

10. The average distance between the Earth and the Sun is defined as the *Astronomical Unit* (or AU), which is equivalent to 1.496×10^8 km. The average distance between the Sun and Jupiter is around 5.2 AU – what is the Jupiter-Sun distance in *kilometers*?

11. The average speed at which the Earth orbits the Sun is around 29.8 km/s. What is the Earth's orbital speed in km/yr? What is the Earth's orbital speed in AU/yr? (1 yr = 3.15×10^7 s).

12. Another common unit system used in astronomy is the centimeter / gram / second (CGS) system, where distances, masses, and time are respectively reported in these units. What is mass of the Earth in this unit system? What is the distance of the Earth to the Sun in this unit system?

4 Parallax

As we've seen in previous problems, the distances we talk about in astronomy are *very large*. You might imagine it's difficult to figure out how far away astronomical objects are from us here on Earth. While this is true for very distant objects, we can actually use basic concepts in trigonometry to figure out how distant some of our neighbors are. The method we use is called **astronomical parallax**.

Here's the basic idea: as the Earth orbits the Sun, our position in space changes over the course of the year. If you look at a nearby star from two different points in Earth's orbit (say, six months apart), that star will appear to have shifted slightly against the background of much more distant stars, which barely appear to move at all (see Figure 1). This apparent shift is called the **parallax angle**, usually written as θ. The farther apart your two observation points are (the **baseline**, x), the bigger this shift appears – and the farther away the star is, the *smaller* the shift appears. You'll feel this yourself in a moment using nothing but your own thumb.

Because real stars are so far away, these angles are extremely small – often much less than one degree. This is actually why parallax stops working for very distant objects: at some point, the angle becomes too tiny for even our best telescopes to measure reliably. Astronomers can only use parallax for relatively nearby stars in our own galaxy.

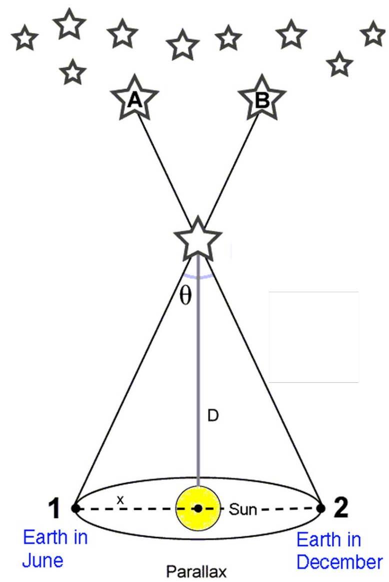


Figure 1: An illustration of parallax. Points 1 and 2 mark Earth's position in its orbit six months apart (in June and December respectively), separated by baseline x, with the Sun at the midpoint. θ is the full parallax angle – the angular shift between the two sightlines to the star's true position. D is the distance from the Sun to the star. Because of this shift in viewing angle, the star appears to be at position B when observed from point 1 (June), and at position A when observed from point 2 (December), relative to the much more distant background stars.

4.1 The length of your arm

Using parallax to calculate distance is a very simple and physical process. All you need to find is the angle that something appears to move against an object at infinity, and the baseline that you move in the process. To illustrate parallax, hold out your hand in a thumbs up position, close one eye, and then switch eyes. Notice how your thumb seems to move against the background. The background here is acting as 'object at infinity'.

Qs: Write down in your lab write-up the following.

13. The equation for D (the distance) in terms of x (the baseline, distance between points 1 and 2 in Figure 1) and θ (the full angle) using Figure 1.

14. How does the angle change when you increase the baseline, x?

15. How about when you increase the distance, D?

Get a protractor for your group. Try this out in the hall with your thumb: Line up 0° on the protractor with one edge of your thumb. With one eye, line up the same edge of your thumb with something in the background. Remember that spot, and then switch eyes. Notice where that same edge of your thumb is relative to the background. Note that the protractor appeared to shift too! Slide the edge of the protractor along your thumb, until the 0° line is lined back up with where the edge of your thumb was originally. Check approximately how many degrees the edge of your thumb moved. Now, measure the distance between your eyes:

16. **Calculate the length of your arm (in cm) using parallax. Does this make sense? Measure your arm using the meter stick. Were you close? Discuss.**

4.2 Distance to nearby stars

Now let's apply parallax to real stars. There's one small but important difference in convention here: instead of using the *full* shift between two extreme points, astronomers define the **parallax angle** p using *half* of the total shift you measured in Section 4.1, observed from *half* of the baseline – specifically, a baseline equal to 1 AU, the distance from the Earth to the Sun (which is half the distance across Earth's full orbit). Astronomers do this because it's convenient: it lets us compare a star's position directly against where the Sun is, rather than against Earth's other orbital position six months later.

This gives us a distance unit called the **parsec** (pc), defined directly from this half-angle convention: a star with a parallax angle of $p = 1$ arcsecond ($1''$), measured using a baseline of 1 AU, is defined to be exactly 1 parsec away. This gives us a very simple relationship between parallax angle and distance:

$$D\,[\text{pc}] = \frac{1}{p\,['']} \tag{1}$$

where p is the parallax angle measured in arcseconds. Notice that this is the same small-angle relation you explored in Section 4.1, just applied to half of the baseline and half of the angle – a bigger parallax angle still means a *closer* star, and a smaller angle still means a *farther* star.

Below are the measured parallax angles for a few well-known, nearby stars:

Star	**Parallax Angle, p ($''$)**
Proxima Centauri	0.768
Sirius	0.379
Vega	0.130
Betelgeuse	0.0055

Qs: Remember to show your work! **Record your solutions in your lab write-up.**

17. Calculate the distance (in parsecs) to each of the four stars listed above.

18. Which star is closest to us? Which is farthest? Does this match your intuition about which parallax angles correspond to closer vs. farther objects?

19. 1 parsec is equivalent to about 3.26 light years. Convert your four distances into light years.

20. Proxima Centauri is the closest known star to the Sun (other than the Sun itself). Does your calculated distance seem reasonable, given that light from the Sun takes about 8 minutes to reach Earth?

21. Betelgeuse has a much smaller parallax angle than the other stars, making it harder to measure precisely. Based on what you found in Section 4.1 (how angle changes with distance), why might astronomers have more difficulty pinpointing the exact distance to farther stars like Betelgeuse?

5 Wrapping things up

Qs: Complete this section by yourself. **Record your responses in your lab write-up**. These questions are graded for attendance/completion they need not be long or super detailed. :)

22. Why is scientific notation useful in astronomy (and in quantitative disciplines in general)? What are "orders of magnitude" and why do we care about them? Furthermore, why is it important to include units?

23. How do you feel about science, math, and astronomy? This can be a very short answer if you want; are you nervous with numbers, or pretty comfortable?

24. (Optional): Is there anything you would like me to know about you? If so please tell me! Also please write any questions, comments, and suggestions you might have about this lab!